K L Deemed to be University	**Department of Electronics and Communication Engineering -- KLVZA** **Course Handout** **2025-2026, Summer Term**

Course Title	**: ASIC AND FPGA DESIGN**
Course Code	**: 24EC2237F**
L-T-P-S Structure	**: 2-0-2-0**
Pre-requisite	: Nil
Credits	**: 3**
Course Coordinator	**: Bukya Balaji**
Team of Instructors	: Bukya Balaji, Bukya Balaji
Teaching Associates	:
Syllabus :	Introduction to ASICs: Types of ASICs, Design Flow, Case study, Economics of ASICs, ASIC Cell Libraries, Combinational Logic Cells, Sequential Logic Cells, Data path Logic Cells, I/O Cells. ASIC Library Design: Library-Cell, Library Architecture, Gate-Array, Standard-Cell, Datapath-Cell Design. Programmable ASICs: The Antifuse, Static RAM, EPROM, and EEPROM Technology. MOS Programmable Logic Device (PLD). Programmable ASIC Logic and I/O cells: Logic Cells: Actel ACT, Xilinx LCA, Altera FLEX, Altera MAX, AMDs – CPLD (Mach 1 to 5), Cypress FLASH 370. I/O Cells: DC Output, AC Output, DC Input, AC Input, Clock Input, Xilinx I/O Block. Programmable ASIC Interconnect and Construction: ASIC Interconnect: Actel ACT, Xilinx LCA, Xilinx EPLD, Altera MAX 5000 and 7000, Altera MAX 9000, Altera FLEX. ASIC Construction: Physical Design, FPGA Partitioning, Partitioning Methods, Floorplanning, Placement, Routing.
Text Books :	1. Michael John Sebastian Smith, “Application Specific Integrated Circuits”, Pearson Education,2001. 2. Debaprasad Das, “VLSI Design”. 3. Neil H. E. Weste and K. Eshraghian, Principles of CMOS VLSI Design, 2nd Edition, Addison Wesley.
Reference Books :	1. Bob Zeidman, “Designing with FPGAs and CPLDs”. 2. Stephen Brown and ZvonkoVranesic “Fundamentals of Digital Logic with Verilog Design”. 3. Pak K. Chan, Samiha Mourad,“Digital Design Using Field Programmable Gate Array”.
Web Links :	http://www.pvpsiddhartha.ac.in/dep_it/lecture%20notes/DSD/unit4.pdf https://www.slideshare.net/yayavaram/introduction-to-asicsntro-asi-cs https://www.slideshare.net/subashjohn1/system-partitioning-and-its-considerationshttps://www.linkedin.com/learning-admin/content/learning-paths/1~AAAAAAVU26I=1010033?account=89447330
MOOCS :	https://www.coursera.org/programs/kl-university-faculty-program-lub63/learn/intro-fpga-design-embedded-systems?source=search.
Course Rationale :	This course offers students a theoretical understanding of ROM, PLA, PAL, Analysis of Clocked Sequential Circuits, ASIC Design, Complex Programmable Logic Devices (CPLDs), Field Programmable Gate Array (FPGA).
Course Objectives :	To prepare the student to be an entry-level industrial standard ASIC or FPGA designer. To give the student an understanding of issues and tools related to ASIC/FPGA design and implementation. To give the student an understanding of basics of System on Chip and platform-based design.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply the knowledge of ASIC types, design flow, and economic aspects to identify appropriate cell libraries (combinational, sequential, datapath, and I/O) for design.	PSO1,PO1	3
CO2	Apply the understanding of ASIC library design and programmable logic technologies (Antifuse, SRAM, EEPROM, etc.) to evaluate suitable logic for implementation.	PO1,PSO1	3
CO3	Analyze and compare logic and I/O cell architectures across various programmable ASIC families like Actel, Xilinx, Altera, AMD, and Cypress.	PO1,PO2,PO4,PSO1	4
CO4	Analyze interconnect techniques, partitioning methods, and physical design steps such as floorplanning, placement, and routing in ASIC construction.	PO2,PSO1,PO4,PO1	4
CO5	Design and implement digital systems using HDL on FPGA/ASIC platforms with simulation, synthesis, and verification using industry tools.	PO1,PO2,PO3,PO5,PO9,PO10,PSO1	5

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5
CO1	3	**Btl-1** Identify and classify the types of ASICs, based on architecture and use case.	**Btl-2** Illustrate and interpret the ASIC design flow; Recognize and categorize different ASIC cell libraries.	**Btl-3** Analyze the cost and economic trade-offs of ASIC design.		
CO2	3	**Btl-1** Interpret the cell library architecture and classification (gate array, standard cell, datapath cell).	**Btl-2** Describe the characteristics of programmable ASICs: Antifuse, SRAM, EPROM, EEPROM.	**Btl-3** Apply the MOS-based PLDs in custom logic design.		
CO3	4	**Btl-1** Identify and classify the architectures of logic cells in	**Btl-2** Compare I/O cell types and functionalities.	**Btl-3** Apply the architecture of programmable logic cells (e.g.,	**Btl-4** Analyze the suitability of ASIC family and cell types	

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5
		Actel, Xilinx, Altera, AMD.		Actel ACT, Xilinx LCA) to design simple digital circuits using vendor-specific features.	for different applications.	
CO4	4	**Btl-1** Describe the knowledge of floorplanning, placement, and routing in ASIC design.	**Btl-2** Compare partitioning strategies and its methods.	**Btl-3** Apply floorplanning and placement concepts to a given digital system block diagram using basic ASIC design principles.	**Btl-4** Analyze ASIC interconnect methods across different vendors (Actel, Xilinx, Altera).	
CO5	5	**Btl-1** Study the conceptsof ASIC modulesusing Verilog.	**Btl-2** Understand the basicconcepts of ASICmodules using Verilog.	**Btl-3** Applying the synthesizable HDL code (Verilog/VHDL) for digital systems.	**Btl-4** Analyzing the simulate and synthesize designs using EDA tools (Vivado, Quartus, etc.).	**Btl-5** Implement HDL designs on FPGA boards and verify correctness

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

Po No.	Program Outcome
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PSO1	To apply concepts of micro and nanoelectronics, embedded systems, and circuit analysis to develop smart electronic systems for real-world applications.

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Introduction to Course Handout and Types of ASICs.	Textbook1 [CH:1] [Page 2]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
2	CO1	COI-2	Design Flow and Case study	Textbook1 [CH:1] [Page 7]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
3	CO1	COI-2	Economics of ASICs and ASIC Cell Libraries	Textbook1 [CH:1] [Page 9]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
4	CO1	COI-2	Combinational Logic Cells	Textbook1 [CH:2][Page 15]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
5	CO1	COI-3	Sequential Logic Cells.	Textbook1 [CH:2][Page 17]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
6	CO1	COI-3	Data path Logic Cells and I/O Cells [Flipped learning]..	Textbook1 [CH:2][Page 35]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
7	CO2	COI-1	Library-Cell and Library Architecture	Textbook1 [CH:3][Page 18]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
8	CO2	COI-2	Gate-Array and Standard-Cell	Textbook1 [CH:3][Page 20]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
9	CO2	COI-2	Data path-Cell Design and Programmable ASICs	Textbook1 [CH:3][Page 29]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
10	CO2	COI-2	The Antifuse and Static RAM	Textbook1 [CH:4][Page 1]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
11	CO2	COI-3	EPROM and EEPROM Technology	Textbook1 [CH:4][Page 4]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
12	CO2	COI-3	MOS Programmable Logic Device (PLD) [Flipped learning].	Textbook1 [CH:4][Page 7]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
13	CO3	COI-1	Programmable ASIC Logic, Logic Cells and Actel ACT	Textbook1 [CH:5][Page 1]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
14	CO3	COI-2	Xilinx LCA, Altera FLEX and Altera MAX	Textbook1 [CH:5][Page 9]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
15	CO3	COI-2	AMDs – CPLD (Mach 1 to 5)	Textbook1 [CH:5][Page 12]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
16	CO3	COI-3	Cypress FLASH 370	Textbook1 [CH:5][Page 14]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
17	CO3	COI-3	I/O Cells, DC Output and AC Output	Textbook1 [CH:6][Page 1]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
18	CO3	COI-4	DC Input and AC Input	Textbook1 [CH:6][Page 12]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
19	CO4	COI-4	Clock Input and Xilinx I/O Block [Flipped learning].	Textbook1 [CH:6][Page 15]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
20	CO4	COI-1	ASIC Interconnect and Actel ACT	Textbook1 [CH:7][Page 1]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
21	CO4	COI-2	Xilinx LCA and Xilinx EPLD	Textbook1 [CH:7][Page 7]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
22	CO4	COI-3	Altera MAX 5000 and 7000	Textbook1 [CH:7][Page 11]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
23	CO4	COI-3	Altera MAX 9000 and Altera FLEX	Textbook1 [CH:7][Page 12]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
24	CO4	COI-4	ASIC Construction, Physical Design and FPGA Partitioning	Textbook2 [CH:11][Page 357]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
25	CO4	COI-4	Partitioning Methods and Floorplanning	Textbook2 [CH:11][Page 358]	Chalk,Talk	End Semester Exam,SEM-EXAM2
26	CO4	COI-4	Placement and Routing [Flipped learning].	Textbook2 [CH:11][Page 372]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Understand Course handout and Types of ASICs.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction to Course	1	Talk	--- NOT APPLICABLE ---
20	Course handout description, discussion on syllabus and evaluation plan	2	Talk	--- NOT APPLICABLE ---
20	Types of ASICs	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Describes about the design flow of ASIC and its case

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Design Flow of ASIC	2	PPT	--- NOT APPLICABLE ---
20	Case study	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Explain about the Economics of ASICs and ASIC Cell Libraries.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Economics of ASICs	2	PPT	One minute paper
20	ASIC Cell Libraries	3	PPT	One minute paper
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 Demonstrate the Combinational Logic Cells.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Introduction of Combinational Logic Circuits	2	PPT	--- NOT APPLICABLE ---
20	Combinational Logic Cells	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Demonstrate the Sequential Logic Cells.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Introduction to sequential circuits	2	PPT	--- NOT APPLICABLE ---
20	Sequential Logic Cells	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 Summarize the Data path Logic Cells and I/O Cells

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Data path Logic Cells [Flipped learning].	2	PPT	Video synthesis
20	I/O Cells [Flipped learning].	3	PPT	Video synthesis
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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SESSION NUMBER : 7

Session Outcome: 1 Explain about the ASIC Library Design

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Library-Cell	2	PPT	--- NOT APPLICABLE ---
20	Library Architecture	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Describes the Gate-Array and Standard-Cell.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Gate-Array	2	PPT	--- NOT APPLICABLE ---
20	Standard-Cell	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Discuss about the Data path-Cell Design and Programmable ASICs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Data path-Cell Design	2	PPT	Just in-time teaching
20	Programmable ASICs	3	PPT	Just in-time teaching
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Explain about the antifuse and static RAM

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Antifuse	2	PPT	--- NOT APPLICABLE ---
20	Static RAM	3	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Summarize the EPROM and EEPROM Technology

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	EPROM	2	PPT	--- NOT APPLICABLE ---
20	EEPROM Technology	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Demonstrate the MOS Programmable Logic Device (PLD)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	MOS Structure [Flipped learning].	2	PPT	Peer Survey
20	MOS Programmable Logic Device (PLD) [Flipped learning].	3	PPT	Peer Survey
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 Explain the Programmable ASIC Logic, Logic Cells and Actel ACT

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Programmable ASIC Logic	3	PPT	--- NOT APPLICABLE ---
20	Logic Cells and Actel ACT	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Discuss the Xilinx LCA, Altera FLEX and Altera MAX

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Xilinx LCA	3	PPT	--- NOT APPLICABLE ---
20	Altera FLEX and Altera MAX	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Discuss the AMDs – CPLD (Mach 1 to 5)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	AMDs – CPLD (Mach 1 to 3)	3	PPT	Group Discussion
20	AMDs – CPLD (Mach 3 to 5)	4	PPT	Group Discussion
5	Conclusion & Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 1 Demonstrate the Cypress FLASH 370

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Introduction to Cypress FLASH 370	3	PPT	--- NOT APPLICABLE ---
20	Cypress FLASH 370	4	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 Demonstrate the DC Output and AC Output

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	DC Output	3	PPT	--- NOT APPLICABLE ---
20	AC Output	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 1 Summarize the DC Input and AC Input

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	DC Input	3	PPT	--- NOT APPLICABLE ---
20	AC Input	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 1 Analyze the Clock Input and Xilinx I/O Block

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Clock Input [Flipped learning].	3	PPT	Focused listing
20	Xilinx I/O Block [Flipped learning].	4	PPT	Focused listing
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 1 Explain about the ASIC Interconnect and Actel ACT

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	ASIC Interconnect	3	PPT	--- NOT APPLICABLE ---
20	Actel ACT	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 Discuss the Xilinx LCA and Xilinx EPLD

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Xilinx LCA	3	PPT	--- NOT APPLICABLE ---
20	Xilinx EPLD	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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SESSION NUMBER : 22

Session Outcome: 1 Demonstrate the Altera MAX 5000 and 7000

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Altera MAX 5000	3	PPT	Quiz/Test Questions
20	Altera MAX 7000	4	PPT	Quiz/Test Questions
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 1 Apply the Altera MAX 9000 and Altera FLEX

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Altera MAX 9000	3	PPT	--- NOT APPLICABLE ---
20	Altera FLEX	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 1 Analyze the ASIC Construction, Physical Design and FPGA Partitioning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Physical Design	3	PPT	--- NOT APPLICABLE ---
20	FPGA Partitioning	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 25

Session Outcome: 1 Analyze the Partitioning Methods and Floorplanning

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Partitioning Methods	3	PPT	--- NOT APPLICABLE ---
20	Floorplanning	4	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 26

Session Outcome: 1 Analyze the Placement and Routing

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap of the Previous Session	1	Talk	--- NOT APPLICABLE ---
20	Placement [Flipped learning].	3	PPT	Case Study
20	Routing [Flipped learning].	4	PPT	Case Study
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Design Standard Cells Using Verilog Code.	CO5
2	Design 4-bit even parity generator and checker Logic using Verilog Code.	CO5
3	Write Verilog Code to implement 3- input combinational logic using 4x1 Multiplexer.	CO5
4	Write Verilog Code to implement 7-segment display.	CO5
5	Write Verilog Code to implement "D" and "T" flip-flop using "JK" flip-flop.	CO5
6	Design Macrocell for Given Boolean Function Using Verilog Code.	CO5
7	Design CLB for Given Truth Table Using Verilog Code.	CO5
8	Implement Moore Sequence Detector Using Verilog Code.	CO5
9	Implement Mealy Sequence generator Using Verilog	CO5
10	Design Gated Clock 4-bit Up/down Synchronous and Asynchronous counter using Verilog Code.	CO5
11	Design 3-bit Random number generator using Verilog Code.	CO5
12	Implement SRAM using Verilog Code.	CO5

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Design Standard Cells Using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Design 4-bit even parity generator and checker Logic using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Write Verilog Code to implement 3- input combinational logic using 4x1 Multiplexer.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 Write Verilog Code to implement 7-segment display.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Write Verilog Code to implement "D" and "T" flip-flop using "JK" flip-flop.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 Design Macrocell for Given Boolean Function Using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 Design CLB for Given Truth Table Using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Implement Moore Sequence Detector Using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Implement Mealy Sequence generator Using Verilog

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Design Gated Clock 4-bit Up/down Synchronous and Asynchronous counter using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Design 3-bit Random number generator using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Implement SRAM using Verilog Code.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introduction	1	Talk	--- NOT APPLICABLE ---
20	Problem Explanation	3	LTC	--- NOT APPLICABLE ---
20	Design & Simulation	4	LTC	--- NOT APPLICABLE ---
50	Viva & Observation	5	LTC	--- NOT APPLICABLE ---
5	Conclusion & Summary	2	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:
NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
2	Weekly Homework Assignments	1	Combinational and Sequential logic circuits	Explain about Combinational and Sequential logic circuits and design the architecture of PLA.	CO1
4	Weekly Homework Assignments	2	Architecture of Field Programmable Gate Arrays	Design the architecture of Field Programmable Gate Arrays and ASIC and explain its design flow.	CO2
6	Weekly Homework Assignments	3	MAX 5000	Design the architecture of CPLD and MAX 5000/7000.	CO3
8	Weekly Homework Assignments	4	Altera –FLEX 8000/10000	With neat sketches, explain and design the architecture of Altera –FLEX 8000/10000.	CO4

COURSE TIME TABLE:

	Hour	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Day	Component									
Mon	Theory	---	---	V-S1	V-S1	---	---	---	--	--
	Tutorial	---	---	--	--	---	---	---	--	--
	Lab	---	---	--	--	---	---	---	V-S1	V-S1
	Skilling	---	---	--	--	---	---	---	--	--
Tue	Theory	---	---	---	---	V-S1	V-S1	---	---	---
	Tutorial	---	---	---	---	--	--	---	---	---
	Lab	---	---	---	---	--	--	---	---	---
	Skilling	---	---	---	---	--	--	---	---	---
Wed	Theory	---	---	---	---	---	---	---	--	--
	Tutorial	---	---	---	---	---	---	---	--	--
	Lab	---	---	---	---	---	---	---	V-S1	V-S1
	Skilling	---	---	---	---	---	---	---	--	--
Thu	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Fri	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--

	Hour	1	2	3	4	5	6	7	8	9
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Sat	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCS

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCS

EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5
End Semester Summative Evaluation Total= 40 %	Lab End Semester Exam	Weightage	16		140					16
		Max Marks	50							50
	End Semester Exam	Weightage	24		180	6	6	6	6	
		Max Marks	100			25	25	25	25	
In Semester Formative Evaluation	Continuous Evaluation - Lab Exercise	Weightage	10		140					10
		Max Marks	50							50

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5
Total= 24 %	**Home Assignment and Textbook**	Weightage	6		60	1.5	1.5	1.5	1.5	
		Max Marks	100			25	25	25	25	
	ALM	Weightage	8		60	2	2	2	2	
		Max Marks	100			25	25	25	25	
In Semester Summative Evaluation Total= 36 %	**Lab In Semester Exam**	Weightage	8		90					8
		Max Marks	50							50
	Semester in Exam-II	Weightage	14		90			7	7	
		Max Marks	50					25	25	
	Semester in Exam-I	Weightage	14		90	7	7			
		Max Marks	50			25	25			

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Bukya Balaji	L	1-MA	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Bukya Balaji	P	1-MA	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(Bukya Balaji)(5155)(ECE)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of ECE

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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